Combating Chirp Interference for Multi-target LoRa Localization

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The long-range and low-power properties of LoRa facilitate its rapid deployment in many location-based services. However, existing LoRa-based localization techniques assume that the received signal is solely from a single node, without any concurrent transmissions from other LoRa nodes. This is because concurrent transmissions lead to mutual interference that inevitably distorts the estimated channel state information (CSI), resulting in significant localization errors. Although interference caused by concurrent transmissions has been studied and addressed in LoRa communication, none of these methods are effective for LoRa localization. This is because localization relies on distinct features and encounters different challenges compared to LoRa communication. To address this fundamental limitation, we propose CLoc, the first LoRa-based multi-target localization method, which is capable of localizing multiple LoRa nodes simultaneously under concurrent transmissions. Through comprehensive analysis, CLoc classifies the interference into two categories based on the chirp slope, i.e., inter-slope interference (different slopes) and co-slope interference (same slope), and identifies their fundamental impacts on CSI errors. CLoc designs dedicated methods that smartly leverage LoRa chirp characteristics to address CSI distortion caused by inter-slope interference, and tackle CSI ambiguity and errors caused by co-slope interference, thereby enabling accurate CSI estimation. We implement the prototype of CLoc with USRP B210 and commodity LoRa nodes. Evaluations under different settings demonstrate that CLoc achieves median localization errors of 3.3 m in a 293,250 m^2 outdoor area and 3.5 m in a 6,750 m^2 indoor area, reducing the localization errors by up to 90.6% compared with the state-of-the-art single LoRa node localization method.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing systems and tools**; • **Networks → Location based services**.

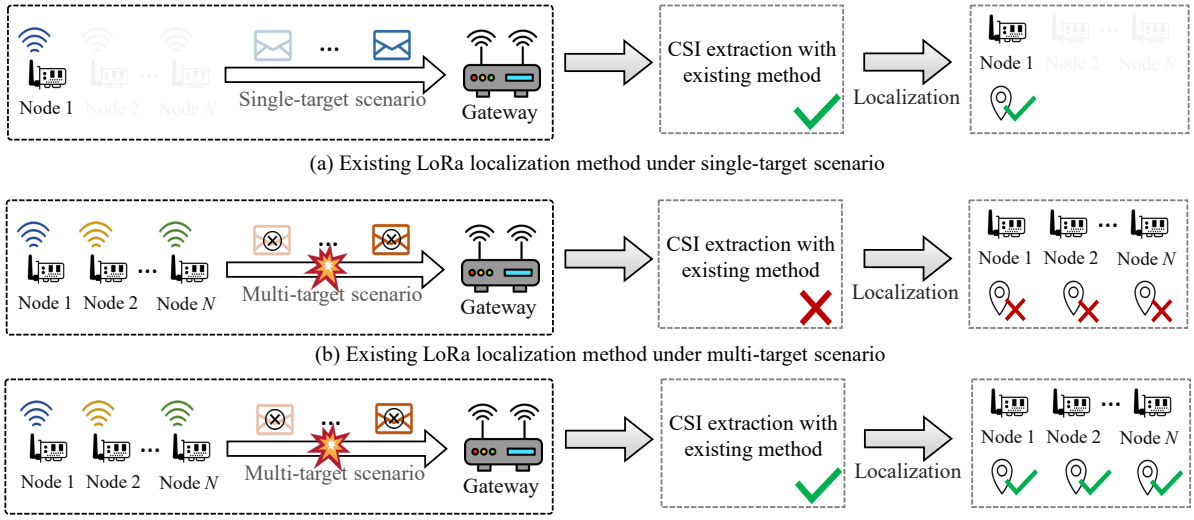
Additional Key Words and Phrases: LoRa, Multi-target localization, Chirp interference, CSI

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(a) Existing LoRa localization method under single-target scenario

(b) Existing LoRa localization method under multi-target scenario

(c) CLoc under multi-target scenario

Fig. 1. Comparison between CLoc and existing LoRa-based localization methods.

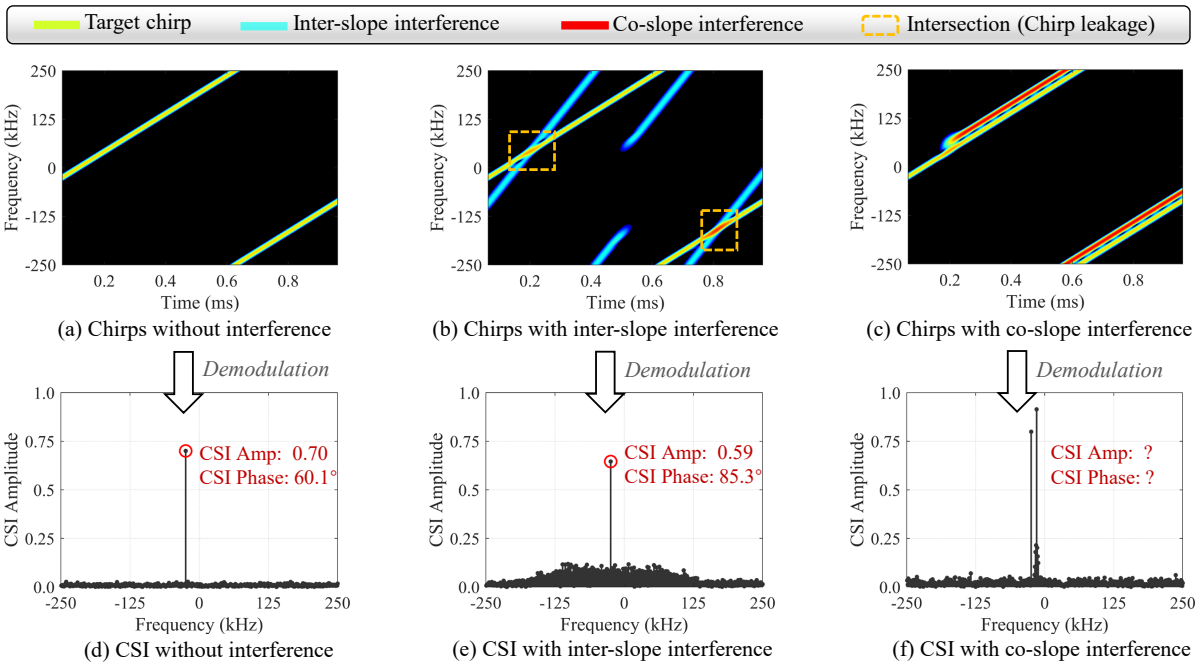
1 INTRODUCTION

Localization serves as the foundation of numerous critical location-based services, including livestock localization [32], personnel search and rescue [30], or asset tracking [38], etc. Although the Global Positioning System (GPS) [43] offers pervasive localization in outdoor environments, it suffers from significant localization errors in indoor environments and also demands high energy consumption, which is not feasible for low-power devices. To address these limitations, researchers have turned to localization via wireless technologies, such as WiFi [17, 56, 65], ZigBee [1, 21], Bluetooth [4, 11, 66], and Ultra-WideBand (UWB) [50, 60]. Although these solutions demonstrate the feasibility of accurate localization, they suffer from a very limited range (e.g., 30 m).

Fortunately, the emerging wireless technology, LoRa, a representative Low-Power Wide-Area Network (LPWAN) technique, complements these limitations by enabling long-range and low-power connectivity. As a result, LoRa presents great potential in enabling long-range and large-scale IoT localization [29]. Recent studies [8, 9, 25, 26, 61] have demonstrated the feasibility of achieving long-range localization with LoRa.

While promising, existing LoRa localization methods assume that only a single LoRa node transmits the signal without interference from other nodes in the LoRa network. Thus, the received signal is clean, and the location of the LoRa node can be estimated accurately [26, 61], as shown in Fig. 1(a). However, given the rapid expansion of the LoRa network with hundreds of nodes connected to a single gateway [41], simultaneous transmissions from multiple LoRa nodes have become increasingly common, leading to severe interference with each other and large localization errors, as shown in Fig. 1(b). For example, in large inventory tracking systems, a high density of LoRa nodes can cause frequent concurrent transmissions, resulting in interference for package localization and tracking. In large-scale agriculture deployments where LoRa is used to monitor livestock location and location-based soil moisture, thousands of LoRa-based sensors operate and report data regularly, greatly increasing the likelihood of multiple LoRa nodes transmitting at the same time. To make LoRa localization adopted in real-life scenarios, we believe one critical requirement is to ensure LoRa localization work in the presence of multiple transmissions.

Although the interference from multiple transmissions has been extensively studied in the field of LoRa communication [10, 12, 14, 22, 39, 46, 51, 54, 55, 62, 63], these existing solutions are not applicable to LoRa-based localization, because of the inherent difference between localization and communication. Specifically, LoRa



(a) Chirps without interference (b) Chirps with inter-slope interference (c) Chirps with co-slope interference

(d) CSI without interference (e) CSI with inter-slope interference (f) CSI with co-slope interference

Fig. 2. Impact of inter-slope interference and co-slope interference on CSI estimation.

communication relies on the features in the frequency domain to distinguish transmissions from multiple LoRa nodes and decode the transmitted data. In contrast, LoRa localization relies on Channel State Information (CSI) to calculate angle-of-arrival (AoA) [26] or generate fingerprints [61]. Our experiments reveal that interference leads to much more severe performance degradation in LoRa localization than in communication. For instance, two LoRa chirps with different slopes lead to severe interference with each other for localization, while this issue causes little impact on communication and is ignored in existing communication literature [54, 62, 63].

In this paper, we investigate the impact of chirp interference[1] on LoRa localization and propose CLoc, the first LoRa-based multi-target localization method that aims to address the interference from concurrent transmissions and accurately localize LoRa nodes simultaneously, as shown in Fig. 1(c). To achieve this objective, CLoc conducts an in-depth analysis on how the interference from other transmissions distorts the estimated CSI, and designs novel techniques to address the interference and extract accurate CSI. With accurate CSI, CLoc calculates the angle of arrival (AoA) of each LoRa node with respect to the LoRa gateways and estimates the location. Note that we localize the LoRa nodes one by one. For each node, its chirp signal is treated as the target chirp, while the chirp signals from other nodes are considered as interference chirps. By iteratively applying this strategy, CLoc effectively calculates the CSI for all the nodes and calculates their locations.

To enable accurate CSI estimation for localizing LoRa nodes, CLoc begins with a thorough analysis of how the interference causes CSI distortion. Since we observe the chirp interference distorts CSI in two different ways, CLoc categorizes these interference into two types based on the chirp slope (i.e., the rate of frequency change): (i) *inter-slope interference*[2] where the interference chirp and the target chirp have different chirp slopes as shown in Fig. 2(b), and (ii) *co-slope interference* where they share the same chirp slope as presented in Fig. 2(c).

[1]We use *chirp interference* and *interference caused by concurrent transmissions* interchangeably in this paper.
[2]Note that in the field of LoRa communication, two chirps with different chirp slopes cause little interference with each other [3, 62, 63]. However, in the domain of LoRa localization, we observe these chirps can cause severe interference that distorts CSI.

Next we explain how these two types of chirp interference distort CSI. **Challenge 1: CSI distortion induced by inter-slope interference.** As shown in Fig. 2(b), the interference chirp (highlighted in blue) has a different chirp slope from the target chirp as highlighted in green, therefore, it intersects with the target chirp in both the time domain and frequency domain. At the intersection parts, the target chirp and interference chirp overlap, causing a portion of power from the interference chirp to leak into the target chirp. As shown in Fig. 2(e), even though we can still identify the CSI of the target chirp after demodulation, it is different from that without interference in Fig. 2(d). As a result, the small amount of leakage can change the CSI of the target chirp and correspondingly cause CSI estimation error. We term this issue as chirp leakage. **Challenge 2: CSI ambiguity and distortion caused by co-slope interference.** Unlike inter-slope interference, co-slope interference occurs if the target chirp and interference chirps share the same slope and exhibit similar time-frequency patterns as shown in Fig. 2(c). After demodulation as shown in Fig. 2(f), it is difficult to identify the CSI of the target chirp because there are multiple peaks and each of them represents one CSI value, leading to the CSI ambiguity issue. Furthermore, we also observe that when the interference chirp is very close to the target chirp in the frequency domain, i.e., two very close peaks in Fig. 2(f), the CSI of the target chirp is changed by the interference chirp, resulting in CSI estimation error.

CLoc proposes novel methods to address these two types of interference. For the *inter-slope interference*, CLoc leverages the property of different chirp slopes to remove the interference chirp from the received signal and eliminate the impact of interference chirp, i.e., chirp leakage, on CSI estimation. While this method removes the interference source, the intersection part shared by both the target chirp and interference chirp is also removed, rendering the remaining target chirp incomplete and causing a reduction in chirp power. To address this effect, CLoc proposes a coherent chirp combination method to coherently combine multiple target chirps within the same LoRa packet to significantly improve the signal power. For the *co-slope interference*, CLoc leverages the CSI characteristics of both the target chirp and the interference chirp over a short time period to address the CSI ambiguity. Specifically, the CSI of target chirps is relatively stable during a short time, while the CSI of interference chirps fluctuates randomly due to the misalignment between the interference chirp and the processing window. Additionally, CLoc adopts a windowing technique to suppress the CSI estimation error caused by the nearby co-slope interference chirps. With the proposed interference removal methods, CLoc can accurately extract the CSI values for all LoRa nodes and then adopt super-resolution AoA estimation to localize them concurrently.

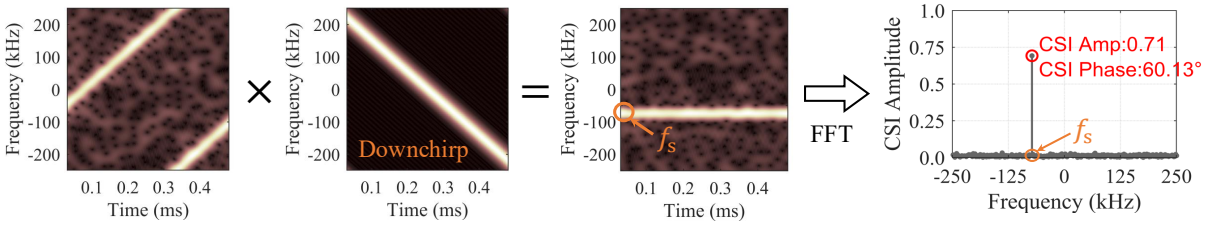
Contributions. The main contributions of this paper are summarized as follows.

- To the best of our knowledge, CLoc is the first LoRa-based multi-target localization method that addresses different types of chirp interference caused by concurrent transmissions.
- CLoc conducts an in-depth analysis on how the concurrent transmissions interfere with the CSI estimation for multi-target localization and categorizes the interference into two types: inter-slope interference and co-slope interference. CLoc designs novel methods to address these interference, enabling accurate CSI estimation and node localization.
- We evaluate CLoc with a real-world testbed. Comprehensive experiment results demonstrate that CLoc is able to localize multiple LoRa nodes with median localization errors of 3.3 m in a 293,250 m^2 outdoor area and 3.5 m in a 6,750 m^2 indoor environment, reducing the localization errors by up to 90.6% compared with the state-of-the-art AoA-based single-target LoRa localization approach.

2 PRELIMINARY

2.1 LoRa Communication

LoRa adopts Chirp Spread Spectrum (CSS) for modulation [3] in the physical layer. It employs a series of upchirps with different initial frequencies to represent LoRa symbols. Specifically, the base unchirp sweeps from $-\frac{BW}{2}$ to $\frac{BW}{2}$, where BW is the chirp bandwidth. CSS changes the initial frequency of the base upchirp to modulate



(a) Dechirping process (b) FFT and CSI extraction

Fig. 3. CSI extraction for LoRa.

different symbols. The frequency of a chirp changes linearly over time at a rate $k = \frac{BW^2}{2^{SF}}$, as shown in Fig. 3(a), where SF is the spreading factor. LoRa typically supports SF ranging from 6 to 12 and BW of 125 kHz, 250 kHz, and 500 kHz [3, 62]. To demodulate each symbol, the receiver multiplies each chirp with a base downchirp whose frequency decreases linearly with time. The multiplication turns the time-varying frequencies of the chirp into a constant frequency. After applying FFT (Fast Fourier Transform), a peak can be generated in the frequency domain as shown in Fig. 3(b). This position of peak (i.e., frequency value) is employed to decode the transmitted LoRa symbol [3].

2.2 Channel State Information Estimation for LoRa Localization

Channel State Information (CSI) is commonly utilized for localization [4, 17, 26, 61] as CSI characterizes the wireless channel between the LoRa node and gateway. With CSI, we can estimate the location of the LoRa node with respect to the gateway [8, 17, 26, 56].

Modeling LoRa signal. Suppose that the LoRa node transmits the modulated chirp signals $T(t)$ to the gateway. In general, the transmitted chirp signal can be represented as

$$T(t) = e^{j2\pi(f_s t + \frac{K}{2}t^2 - \frac{BW}{2}t)}, \tag{1}$$

where f_s is the initial frequency of the modulated LoRa chirp, and K is the chirp slope [54]. The transmitted signal propagates via the channel before arriving at the LoRa gateway. The received chirp signal $R(t)$ at the gateway can be represented as

$$R(t) = \underbrace{e^{j2\pi(f_s t + \frac{K}{2}t^2 - \frac{BW}{2}t)}}_{\text{Chirp}} \cdot \underbrace{H_{csi}}_{\text{CSI}}, \tag{2}$$

where H_{csi} represents CSI, which is a complex value.

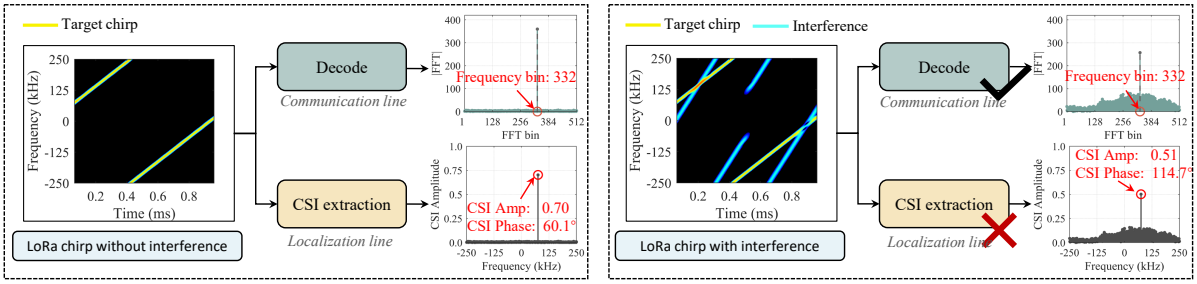
LoRa CSI extraction. Recent techniques leverage the LoRa chirp property to extract CSI. Specifically, as shown in Fig. 3(a), the received signal $R(t)$ is multiplied with a base downchirp $e^{j2\pi(-\frac{K}{2}t^2 + \frac{BW}{2}t)}$ which has an opposite chirp slope as the transmitted upchirp, to turn the time-varying frequencies into a constant frequency f_s.

$$R_{dechirp}(t) = R(t) \cdot e^{j2\pi(-\frac{K}{2}t^2 + \frac{BW}{2}t)} = e^{j2\pi f_s t} \cdot H_{csi}. \tag{3}$$

As presented in Eq. (3), all the signal samples have the same frequency, i.e., f_s. This process is termed as dechirping. After performing FFT as shown in Fig. 3(b), the power of these samples can be accumulated at the frequency of f_s,

$$R_{accu}(f_s) = N \cdot H_{csi}, \tag{4}$$

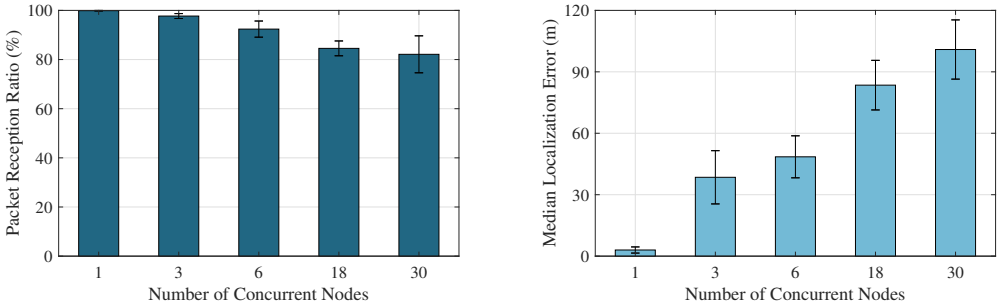
where N represents the number of signal samples within the chirp, which is constant and known in advance. $R_{accu}(f_s)$ is the accumulated signal at the frequency of f_s. During this process, the signal samples are constructively added up while the noise is randomly combined, therefore, $R_{accu}(f_s)$ has a high SNR. The CSI is calculated by the



(a) Decode and CSI extraction without interference

(b) Decode and CSI extraction with interference

Fig. 4. Comparison of LoRa communication (decode) and LoRa localization (CSI extraction) with and without interference.



(a) Performance of LoRa concurrent communication

(b) Performance of LoRa concurrent localization

Fig. 5. Communication performance vs localization performance under concurrent transmissions.

division between $R_{accu}(f_s)$ and N. As shown in Fig. 3(b), by identifying the peak in the FFT spectrum, we can extract the CSI amplitude, and the CSI phase can also be obtained in a similar way.

2.3 MUSIC Algorithm for Localization

For LoRa localization, we adopt the angle of arrival (AoA)-based triangulation [8, 26]. The Multiple Signal Classification (MUSIC) algorithm is a super-resolution technique that leverages CSI to estimate the AoA of the transmitter [34]. It has been well studied to resolve the multipath effect efficiently in many wireless localization systems [17, 26, 56]. Specifically, it first leverages the CSI collected by an antenna array to compute a covariance matrix. Then, it performs eigenvalue decomposition on the covariance matrix to get two sets of eigenvectors corresponding to the signal subspace and noise subspace. MUSIC utilizes the orthogonality between the signal and noise subspace to construct a spatial spectrum and identifies multiple peaks in the spectrum, corresponding to multiple estimated AoA values because of the multipath.

3 MOTIVATION

LoRa is promising for long-range wide-area IoT localization. Existing LoRa localization methods assume that there is just a single LoRa node transmitting the signal to the gateway, so that the received signal at the gateway is clean and the location of the LoRa node can be accurately estimated. However, in LoRa networks, thousands of nodes can be connected to a single gateway, resulting in overlapped transmissions and mutual interference among these nodes [47, 54]. We observe that these interference can cause significant localization errors, hindering the adoption of LoRa localization in real-life scenarios. In addition, concurrent LoRa localization can benefit many

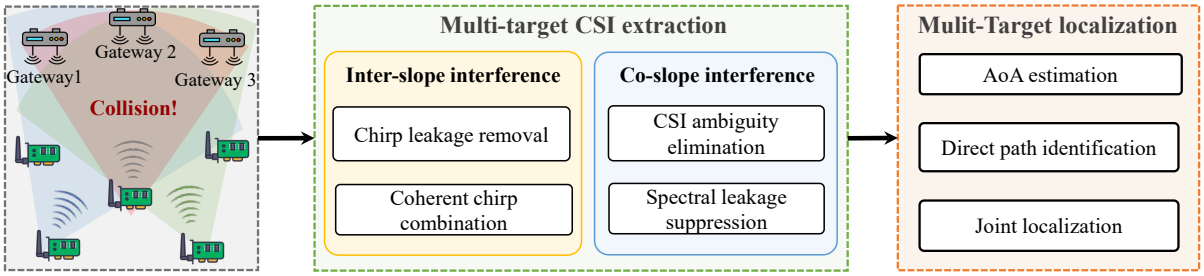


Fig. 6. System overview of CLoc.

applications, such as package localization in large inventory tracking systems and livestock location monitoring in large-scale smart agriculture.

Even though the interference issue caused by concurrent transmissions (i.e., collision) has been well studied in LoRa communication, these solutions are not applicable to LoRa localization. This is because the feature (i.e., CSI) utilized for LoRa localization differs from that (i.e., chirp frequency) in LoRa communication, and we observe that CSI is more vulnerable to interference. Specifically, as shown in Fig. 4(a) and (b), LoRa communication employs the chirp frequency to decode data (upper part of figures), and this frequency does not change with and without interference. However, the CSI utilized for LoRa localization (lower part of figures) is changed in the presence of interference chirps. In Fig. 4(b), the CSI phase error caused by chirp interference can be as large as 54.6°. This is because the impact of chirp interference is determined by the power of the interference chirp and the number of intersections. In Fig. 4(b), the power of the interference chirp is large (i.e., 9 dB higher than that of the target chirp), and the number of intersection parts is 2, resulting in a large CSI phase error.

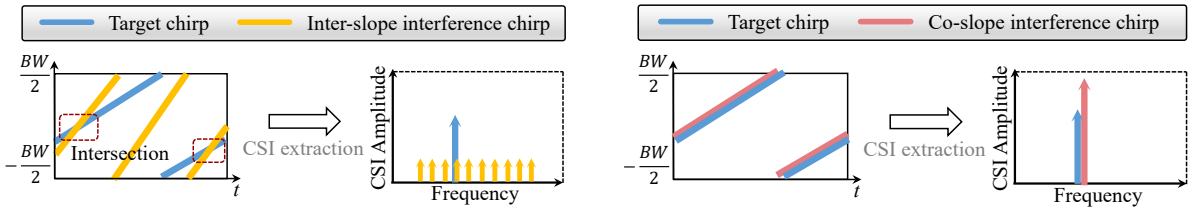
We further conduct preliminary experiments to compare the performance of LoRa communication and localization in the presence of concurrent transmissions, i.e., interference. We randomly deploy 30 LoRa nodes and 3 gateways in a 293,250 m^2 open park. The LoRa standard typically supports 21 types of chirps with different Spreading Factor (SF) and Bandwidth (BW) configurations [62]. Each LoRa node is assigned a specific parameter configuration according to its distance from the gateway. A larger SF and a narrower BW are generally chosen for a longer distance between the node and the gateway. For LoRa communication, we adopt the state-of-the-art concurrent communication method [54] to address the interference and calculate the chirp frequency for decoding data. As shown in Fig. 5(a), the packet reception ratio (PRR) slightly decreases with the increase of concurrent LoRa nodes, and the PRR is still above 80% even with 30 concurrent LoRa nodes. In contrast, when we adopt the same method to address the interference for localization, the localization errors are very large, as presented in Fig. 5(b). The median localization error is 3.2 m if there is a single LoRa node, but it increases to 39.5 m and 50.7 m when there are three and six LoRa nodes transmitting signals to the gateway, respectively.

This motivates us to design a multi-target LoRa localization method to resolve the impact of concurrent transmissions in the LoRa network.

4 SYSTEM OVERVIEW

Fig. 6 shows the overview of CLoc, which consists of multi-target CSI extraction and multi-target localization.

Multi-target CSI extraction. CLoc aims to address the impact of interference caused by concurrent LoRa transmissions and accurately estimate CSI for each LoRa node. For each node, its chirp signal is treated as the target chirp, while the signals from other nodes are considered as interference chirps. By iteratively applying this strategy, CLoc effectively calculates the CSI for all the nodes. Since we observe the chirp interference distorts CSI in two different ways, we categorize the interference into types based on the chirp slope (i.e., the rate of frequency change): (i) *inter-slope interference* where the interference chirp and the target chirp have different chirp slopes,



(a) With inter-slope interference chirp

(b) With co-slope interference chirp

Fig. 7. Illustration example of target chirp and interference chirp.

and (ii) *co-slope interference* where they share the same chirp slope. To address the inter-slope interference, the LoRa gateway first removes the intersection parts between the target chirp and the interference chirp, where the interference chirp leaks power to the target chirp. However, this process can result in an incomplete target chirp and a power loss issue, making it challenging to extract CSI in low-SNR scenarios. Therefore, we propose a coherent chirp combination to enable accurate CSI estimation even with low SNR. To address the co-slope interference, CLoc eliminates the CSI ambiguity and also addresses the spectral leakage that occurs when the interference chirp is very close to the target chirp in the frequency domain.

Multi-target localization. With CSI, CLoc calculates the angle of arrival (AoA) of each LoRa node with respect to each gateway. By combining the AoA estimations from multiple (e.g., 3) LoRa gateways, CLoc is able to derive the location of the LoRa nodes.

5 MULTI-TARGET CSI EXTRACTION

In this section, we first introduce the two types of interference and how they distort the CSI estimation, and then explain our proposed methods to address these interference.

5.1 Understanding the impact of chirp interference on CSI

As we observe that the chirp interference distorts the CSI in two different ways, CLoc classifies the interference into two types based on the chirp slope: inter-slope interference and co-slope interference.

Inter-slope interference. Inter-slope interference means the interference chirp has different slopes compared with the target chirp. Different from LoRa communication, where chirps with different slopes do not cause decoding errors [62], they do cause significant errors in CSI estimation for LoRa localization. Fig. 7(a) shows an illustration example of the target chirp and an inter-slope interference chirp. The interference chirp intersects with the target chirp, and the intersection part is highlighted with red rectangles. In the intersection part, the interference chirp and target chirp overlap in the time domain and frequency domain. Thus, it is difficult to separate them. When we directly calculate the CSI of the target chirp without interference removal, the target chirp is accumulated at one frequency bin while the interference chirp spreads across a large frequency range, as shown in Fig. 7(a). However, the signal samples of the interference chirp in the intersection part unintentionally accumulate with the target chirp, resulting in the interference chirp leaking power into the target chirp, i.e., chirp leakage. We observe that even though the chirp leakage does not affect the frequency (i.e., the position of the peak) of the target chirp, it significantly impacts the CSI amplitude and phase of the target chirp. This is because the CSI amplitude and phase are highly sensitive to interference, and the small amount of chirp leakage from interference chirps can still distort the CSI of the target chirp. Unlike the CSI amplitude and phase, frequency remains more stable and can be accurately estimated even in the presence of small chirp leakage.

Co-slope interference. We denote the interference as co-slope interference if the interference chirp and the target chirp share the same slope. Fig. 7(b) shows an illustration example of the target chirp and a co-slope

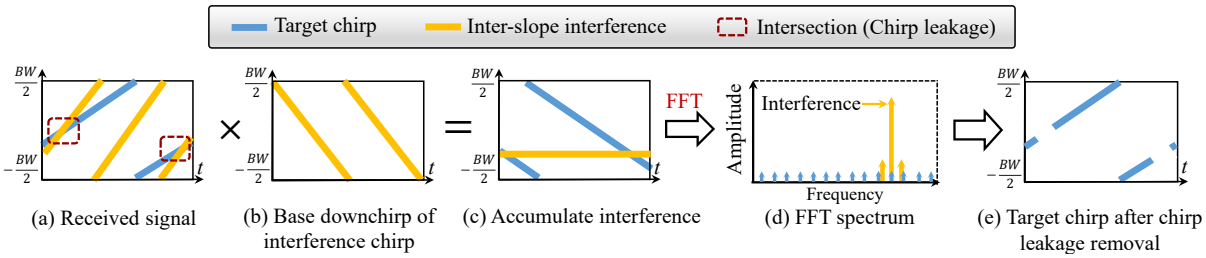


Fig. 8. Chirp leakage removal process.

interference chirp. When we directly calculate the CSI of the target chirp without interference removal, the power of both the target chirp and interference chirp are accumulated, leading to multiple peaks as shown in Fig. 7(b). It is difficult to identify the correct peak for the target chirp's CSI, and we term this problem as the CSI ambiguity issue. Moreover, if the position of the interference peak is close to that of the target peak in the frequency domain, i.e., smaller than 10 frequency bins, the interference chirp also changes the CSI of the target chirp. This problem is well known as spectral leakage [53], which means the energy of one frequency bin leaks into the nearby frequencies.

5.2 Addressing Inter-Slope Interference

In this section, we explain how to address the inter-slope interference. Our solution consists of two modules: chirp leakage removal and coherent chirp combination.

- Chirp leakage removal. The inter-slope interference intersects with the target chirp, as shown in Fig. 8(a). This leads to the chirp leakage issue, which significantly affects the CSI accuracy of the target chirp. This motivates us to remove the source of chirp leakage, i.e., interference chirp (Section 5.2.1).
- Coherent chirp combination. After chirp leakage removal, the target chirp is incomplete as shown in Fig. 8(e). This is because by eliminating the interference chirp, the signal samples of the target chirp at the intersection part are also removed, leading to the target chirp experiencing a power loss issue. This power loss issue degrades SNR and aggravates with the increase of the number of inter-slope interference chirps, i.e., the number of concurrent LoRa transmissions. To address this issue, CLoc adopts the coherent chirp combination that fuses multiple target chirps from the same LoRa packet to ensure accurate CSI estimation in low-SNR scenarios (Section 5.2.2).

5.2.1 Chirp leakage removal. We propose to remove the source of chirp leakage, i.e., the interference chirp, to address its impact on the CSI estimation of the target chirp.

To achieve this objective, CLoc leverages the property of different chirp slopes between the target chirp and interference chirp. Fig. 8(a) shows an example of inter-slope interference where the interference chirp has a different slope from the target chirp. CLoc first accumulates this inter-slope interference chirp in the frequency domain. Specifically, CLoc multiplies the received signal (includes both the interference chirp and the target chirp) with a base downchirp in Fig. 8(b). The slope of the base downchirp is opposite to that of the interference chirp. After applying FFT, the power of inter-slope interference chirp is accumulated at a specific frequency bin with an FFT peak as shown in Fig. 8(d). Meanwhile, the power of the target chirp is distributed across the whole frequency spectrum. This difference motivates us to remove the interference chirp by detecting and eliminating the FFT peaks where the interference chirp is accumulated. We then reverse the two operations (i.e., chirp multiplication and FFT) to recover the original target chirp, by performing IFFT (Inverse Fast Fourier Transform) and applying the ratio between the scattered target chirp and the base downchirp.

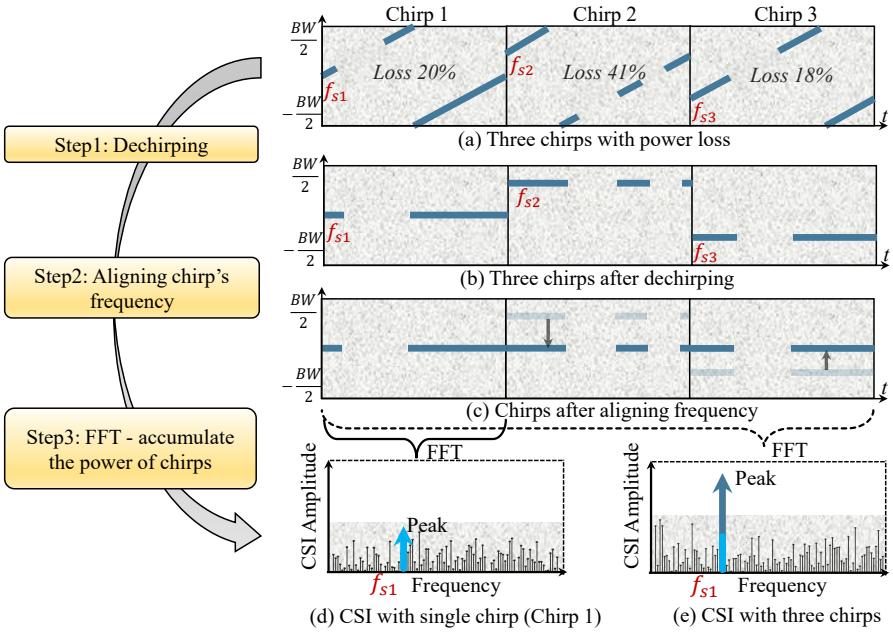


Fig. 9. Addressing power loss issue via coherently combining multiple chirps.

Power loss issue. After chirp leakage removal, the target chirp is incomplete, as shown in Fig. 8(e). This is because the target chirp and the inter-slope interference chirp share the same time and frequency in the intersection area. When the interference chirp is removed, the signal samples of the target chirp in the intersection area are also removed, leading to power loss. This power loss issue is equivalent to the decrease of SNR, making it more challenging to extract accurate CSI in low-SNR scenarios. What is worse, the amount of power loss can vary as shown in Fig. 9(a). This is because the chirp used to modulate data varies, causing the number and locations of intersection areas at the target chirp to vary. The random power loss can corrupt the CSI, which is critical for LoRa localization.

5.2.2 Coherent chirp combination. To address the power loss issue described above, we adopt the coherent chirp combination method. Our basic idea is presented below. A LoRa packet consists of multiple chirps. Even though an individual target chirp suffers from power loss, we can jointly combine multiple chirps within the same LoRa packet to compensate for the power loss. This is because the CSI is relatively stable over a short period and a chirp lasts just a few milliseconds.

The details of our method are explained below. Since each LoRa chirp represents a unique transmitted symbol with specific frequencies, CLoc must align the frequencies of multiple LoRa chirps before coherently combining them. To achieve this, CLoc first adopts LoRa's standard dechirping operation to convert each chirp into a constant frequency signal, as shown in Fig. 9(b). This dechirping operation is achieved by the multiplication between each target chirp and the base downchirp. The chirp signal after dechirping can be represented as

$$R_{dechirp_i}(t) = e^{j2\pi f_{si} t} \cdot H_{csi}, \tag{5}$$

where i represents the target chirp index within the LoRa packet, f_{si} means the starting frequency of ith chirp, and H_{csi} is the desired CSI. As shown in Fig. 9(a), these chirps have different frequencies, which should be aligned before coherently combining them.

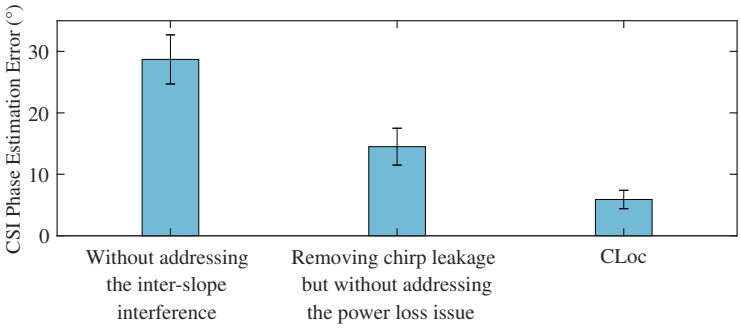


Fig. 10. Benchmark experiment of addressing inter-slope interference.

Then, we propose to align the frequencies of these target chirps. Without loss of generality, CLoc chooses the first chirp as the reference and aligns the frequencies of all other subsequent chirps to the first chirp's frequency. For the ith chirp, its frequency difference to the first chirp is denoted as Δf_i, where $i \neq 1$. To achieve frequency alignment, CLoc multiplies $R_{dechirp_i}(t)$ by $e^{j2\pi\Delta f_i t}$,

$$R_{aligned_i}(t) = R_{dechirp_i}(t) \cdot e^{j2\pi\Delta f_i t}, \tag{6}$$

As a result, all chirps have been tuned to the same frequency, laying the foundation for further chirp combination. Fig. 9(c) shows the chirps after frequency alignment. Note that we can accurately obtain the frequencies of these chirps and their frequency differences via demodulation before dealing with the inter-slope interference (i.e., chirp leakage issue and power loss issue).

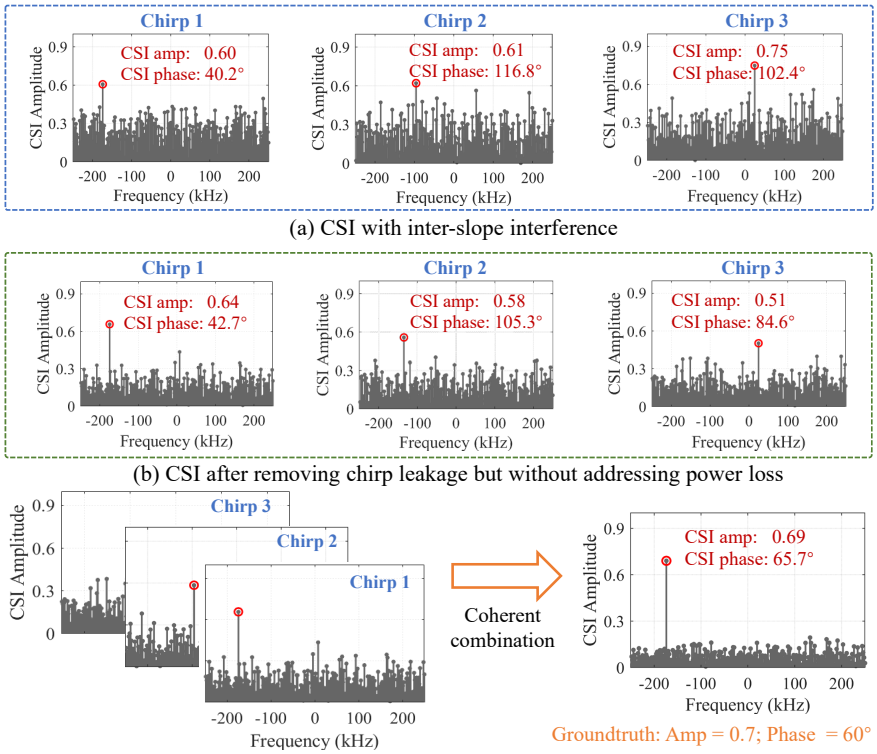
After that, CLoc adopts FFT to accumulate the power of multiple chirps to f_{s1}, i.e., the frequency of the first chirp. As shown in Fig. 9(e), the FFT peak is significantly improved via the combination of three chirps and surpasses the noise floor compared to the FFT peak of a single chirp shown in Fig. 9(d), demonstrating the effectiveness of our method.

Finally, CLoc extracts the CSI amplitude and phase based on the FFT peak. It is important to note that the number of chirps combined plays a crucial role in the performance of coherent chirp combination. The number of chirps combined involves a trade-off between several key factors, including SNR, interference degree, coherence time, and chirp duration. In our evaluation, we account for these factors to determine and adjust the number of chirps that can be effectively combined.

5.2.3 Benchmark experiment. To demonstrate the effectiveness of our method in dealing with inter-slope interference, we conduct benchmark experiments with one target node and three concurrent interference nodes. The target node's spreading factor (SF) and bandwidth (BW) are set to 8 and 500 kHz. The parameter (SF, BW) configurations of the interference nodes are (7, 500 kHz), (9, 500 kHz), and (10, 500 kHz), respectively, to generate interference chirps with different slopes from the target chirps.

We compare CLoc with two baselines: the first baseline directly extracts CSI without addressing inter-slope interference, and the second baseline removes chirp leakage caused by inter-slope interference but does not address the power loss issue. Fig. 10 shows the median error of the estimated CSI phase. Without addressing the inter-slope interference, the CSI phase estimation error is large, with an error of 28.8°. After removing the chirp leakage, the CSI phase estimation error remains high because it still suffers from low SNR caused by the power loss issue. In contrast, CLoc further employs a coherent chirp combination strategy to address the power loss issue, achieving accurate CSI extraction with a phase estimation error of 5.9°.

Fig. 11 shows the process of addressing inter-slope interference. The groundtruth of the CSI amplitude and phase are 0.7 and 60°, respectively. Fig. 11(a) demonstrates that the inter-slope interference severely affects the



(a) CSI with inter-slope interference

(b) CSI after removing chirp leakage but without addressing power loss

(c) CSI after coherently combining three chirps

Fig. 11. An example of CSI extraction under inter-slope interference.

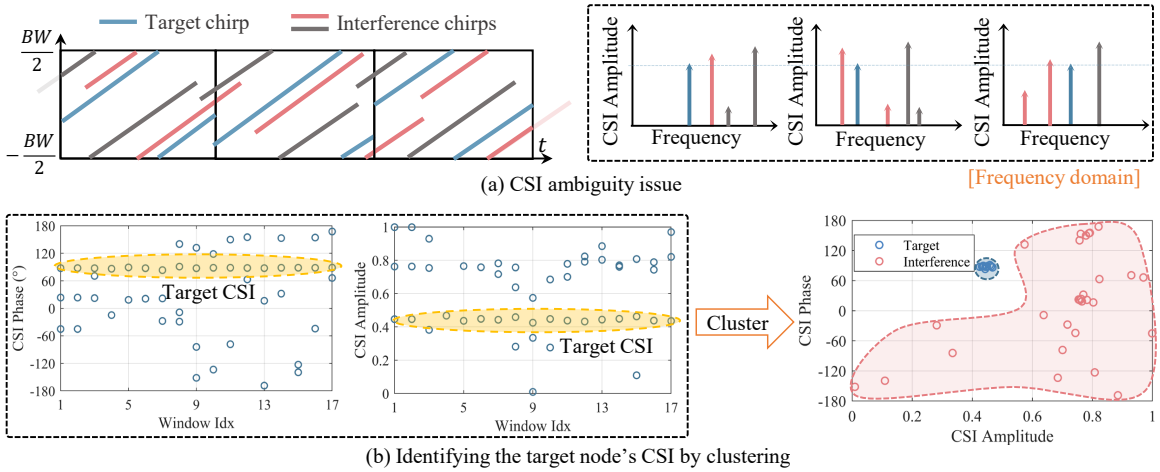
CSI extraction of each chirp. As shown in Fig. 11(b), the proposed chirp leakage removal manages to improve the accuracy of estimated CSI for each chirp but still suffers from low SNR. Specifically, the CSI phase for each chirp still exhibits large deviations from the ground truth, with the errors of 17.3°, 45.3°, and 24.6°. The proposed coherent chirp combination adds up the power of three chirps coherently while the noise remains randomly distributed, thereby improving the SNR and ultimately enabling more accurate CSI extraction. As shown in Fig. 11(c), the phase error of CLoc is significantly reduced to just 5.7° after coherent chirp combination.

5.3 Addressing Co-Slope Interference

In this section, we explain how CLoc addresses the co-slope interference, where the interference chirp and target chirp share the same chirp slope. Our method consists of two modules: CSI ambiguity elimination and spectral leakage suppression.

5.3.1 CSI ambiguity elimination. In the presence of co-slope interference, when extracting CSI, we obtain multiple FFT peaks where each peak's CSI corresponds to a LoRa node. This leads to CSI ambiguity issue and it is difficult to match the CSI value to the correct LoRa node.

To address this CSI ambiguity issue, CLoc takes advantage of the unique CSI stability of the target node. Specifically, we observe that the target node's CSI amplitude (i.e., the amplitude of the FFT peak) is stable with time, while that of the interference node changes with time. The same observation also applies to the target node's CSI phase. As shown in Fig. 12(a), in each processing window, CLoc performs the dechirping and FFT,



(a) CSI ambiguity issue

(b) Identifying the target node's CSI by clustering

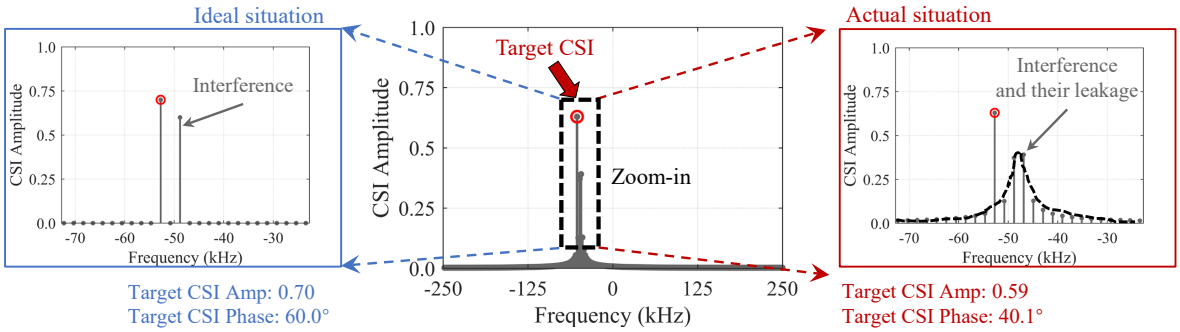
Fig. 12. Addressing CSI ambiguity via clustering.

and there are multiple peaks in the FFT spectrum for each processing window. Across three adjacent processing windows, the target node's CSI amplitudes (highlighted in blue) remain stable, in contrast to the amplitude fluctuations of co-slope interference chirps (highlighted in red and gray). The target node's CSI values resemble each other since the channel state remains stable within a short duration. For the co-slope interference chirp, there is a window misalignment between its actual window and the current processing window of the target chirp, since two LoRa packets commonly do not have the exact starting point. This window misalignment leads to fluctuating CSI amplitude values and phase values for co-slope interference.

Based on the observation described above, CLoc adopts a clustering method to identify the target node's peaks for correct CSI estimation. Our basic idea is to cluster the CSI amplitude and phase of all the peaks and then identify the most stable CSI cluster. Specifically, CLoc first extracts the amplitude and phase of each peak (i.e., CSI) and applies the DBSCAN clustering algorithm [35] to cluster both the amplitude and phase. Fig. 12(b) shows an example of visualizing the target node's amplitude and phase and grouping the dense points. We can observe that the target node's CSI cluster deviates from the CSI points calculated from interference. CLoc adopts DBSCAN for clustering since it can form clusters based on the natural structure of data without having to pre-set the number of clusters. Next, CLoc measures the compactness of each cluster, which is the average distance of each point within a cluster to this cluster center. CLoc selects the most compact cluster to identify the target CSI cluster. As shown in Fig. 12(b), the cluster highlighted in blue corresponds to the target CSI, demonstrating that CLoc can resolve the ambiguity issue and extract the target CSI.

Note that even though the target chirp and interference chirp in our example share the same parameter (i.e., SF and BW) configuration, our solution is still effective even when the co-slope interference chirp has different SF and BW from the target chirp. This is because the co-slope interference chirp with a different SF and BW from the target chirp causes the same issue, i.e., CSI ambiguity. Moreover, these two types of co-slope interference exhibit the same underlying phenomenon: the target CSI remains stable over a short duration, while the interference CSI fluctuates with time.

5.3.2 Spectral leakage suppression. Ideally, one co-slope interference chirp only leads to just one FFT peak in the frequency domain as shown in Fig. 13(a), which could be removed with ambiguity elimination. However, in practice, one co-slope interference chirp leads to multiple adjacent peaks in the frequency domain as demonstrated in Fig. 13(b). This phenomenon is known as spectral leakage [53]. Since there is a time misalignment between the



(a) Ideal CSI without spectral leakage

(b) CSI with spectral leakage

Fig. 13. CSI estimation error induced by spectral leakage.

interference chirp and the processing window, the interference chirp has signal discontinuities at the window boundary, leading to its energy spread into adjacent frequency bins [28, 45]. Our experiment results indicate that if the target peak and interference peak are very close in the frequency domain as shown in Fig. 13(b), spectral leakage from the interference can introduce a large power leakage to the target frequency bins, thus changing the amplitude and phase of target CSI.

CLoc adopts a window function to address the CSI distortion caused by spectral leakage. Specifically, during the demodulation process, CLoc multiplies the received signal with a Hamming window before the FFT operation [45, 57]. Our experimental results show that this method can effectively reduce the spectrum leakage impact of interference, minimizing the number of affected adjacent frequency bins from 10 to 2. This means that if the target peak and interference peak have a distance beyond 2 frequency bins, the co-slope interference has a negligible impact on CSI estimation. Otherwise, if the target peak and interference peak are close, i.e., ±2 frequency bins, the interference still affects the accuracy of CSI estimation. However, the likelihood that the target and interference peaks are within 2 frequency bins is very low.

5.3.3 Benchmark experiment. To illustrate the effectiveness of our method in addressing co-slope interference, we conduct benchmark experiments to validate two key parts, i.e., CSI ambiguity elimination and spectral leakage suppression. In the experiment, we use one target node and three concurrent interference nodes. The spreading factor (SF) and bandwidth (BW) of the target chirp are set to 10 and 250 kHz, respectively. The (SF, BW) configurations of interference chirps are (10, 250 kHz), (10, 250 kHz), and (8, 125 kHz) to generate interference chirps with the same slope as the target chirp. We compare CLoc with two baselines: the first baseline does not address CSI ambiguity and spectral leakage caused by co-slope interference, while the second baseline eliminates CSI ambiguity but does not suppress spectral leakage. The experiment results are shown in Fig. 14. We can observe that the CSI phase estimation error is very large if we do not address the CSI ambiguity issue and spectral leakage. By eliminating the CSI ambiguity, we reduce the CSI phase estimation error to 11.5°. CLoc further reduces the CSI phase error to 6.2°.

5.4 Putting All Together

We summarize the workflow of CLoc's multi-target CSI extraction. The process pipeline involves three steps:

(1) Packet detection. To accurately identify the start of the target chirp in the presence of interference, we employ the correlation method proposed in CIC [39], which addresses the impact of interference on preamble detection. The basic idea is to detect the start time of SFD (Start Frame Delimiter) downchirps in the standard LoRa preamble. Specifically, we correlate the received chirp signals with a standard base downchirp and identify two consecutive peaks that correspond to the SFD downchirps.

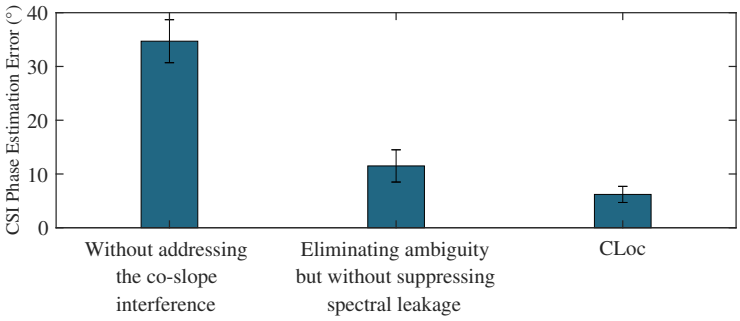


Fig. 14. Benchmark experiment of addressing co-slope interference.

(2) Offset calibration. The transmitter and receiver are not perfectly synchronized in time and frequency, so the received LoRa signal contains carrier frequency offset (CFO) and sampling time offset (STO). CLoc adopts the method proposed in PCube [54] to calibrate these offsets for accurate CSI estimation. Additionally, as discussed in Section 5.2.2, multiple chirps within the same LoRa packet share the same time offset and frequency offset. This allows for the alignment and coherent combination of the chirps.

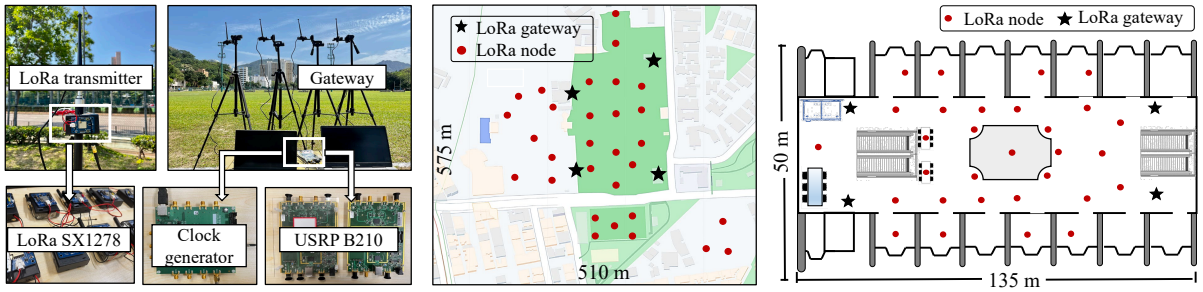
(3) Multi-target CSI extraction. Without loss of generality, CLoc selects one LoRa node as the target node each time and its transmitted chirp as the target chirp, while chirps from other nodes are considered as interference chirps. Then, CLoc identifies the type of interference chirp and applies the corresponding method to address CSI distortion for accurate localization. We repeat this procedure until the CSI of all LoRa nodes within the network is accurately extracted.

6 MULTI-TARGET LOCALIZATION

After performing the multi-target CSI extraction, CLoc obtains accurate CSI measurements for each LoRa node. Base on the CSI measurements, each LoRa gateway estimates the angle of arrival (AoA) and collaborates with other gateways to localize each LoRa node.

AoA estimation. CLoc adopts the super-resolution MUSIC algorithm for AoA estimation [17, 34, 56]. Specifically, it first leverages the CSI measurements collected by an antenna array and computes a covariance matrix. Next, it performs eigenvalue decomposition on the covariance matrix to obtain eigenvectors that represent the signal and noise subspaces. Then, the MUSIC algorithm leverages the orthogonality between these subspaces to construct a spatial spectrum. This is achieved by scanning through all possible AoA directions and assessing how well each direction aligns with the noise subspace. In this spectrum, peaks appear at angles where significant signal energy is detected, corresponding to the estimated AoA values of the incoming signals. Moreover, we can identify multiple peaks in the spectrum, each representing estimated AoA values of different signal paths due to multipath propagation.

Direct path identification. The peaks in the AoA spectrum represent the angles of arrival of multiple paths at the current gateway. One of these peaks corresponds to the direct path between the node and gateway, which is important for localization, while the others are reflected paths from the ground or nearby objects. In general, we utilize the amplitude of the peak to identify the direct path, i.e., the peak with the highest amplitude corresponds to the AoA of the direct path. However, we have observed that the amplitude of the peak associated with reflected paths can surpass that of the direct path. This phenomenon can lead to significant errors in AoA estimation, as the algorithm may incorrectly identify a reflected path as the direct path. Inspired by ArrayTrack [56], CLoc exploits the stability of the AoA spectrum to identify the direct path. The basic idea is that small changes, either from target movement or environmental changes (e.g., a passerby), can dramatically alter reflected paths and



(a) Experimental devices (b) Outdoor scenario (c) Indoor scenario

Fig. 15. Experimental devices and experimental scenarios.

their AoA peaks, while the direct path remains more stable. By leveraging this characteristic, CLoc identifies the direct path's AoA.

Joint localization. To localize each LoRa node, CLoc integrates AoA estimations from multiple (e.g., 3) LoRa gateways. Each node's location is determined by solving a maximum likelihood estimation problem,

$$\hat{l} = \arg\min \sum_{p=1}^{P} \left(\theta_p - L_p(l)\right)^2, \tag{7}$$

where $l = (x, y)$ denotes the candidate location. The estimated target location, denoted as $\hat{l}$, is one location that minimizes the difference between the estimated AoA and the angles calculated from the candidate location. P represents the number of gateways. The location of pth gateway is known and denoted as (x_p, y_p). θ_p is the estimated AoA at the pth gateway. $L_p(l) = \tan^{-1}\frac{y-y_p}{x-x_p}$ calculates the angle between the candidate location (x, y) and the pth gateway. We can use the grid search method [56] to search for the most likely location of the target.

7 EVALUATION

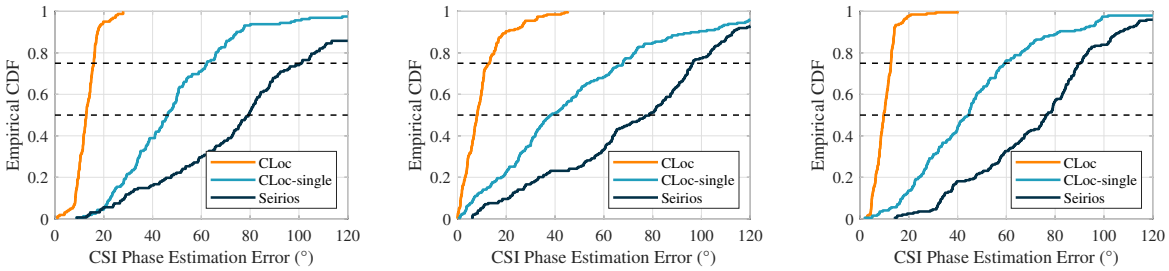
In this section, we evaluate the performance of CLoc with a real testbed under different scenarios and settings.

7.1 Methodology

Experiment setup. We implement the prototype using USRP-based LoRa gateways and commodity LoRa nodes. A LoRa node consists of an SX1278 LoRa chip [36] and an STM32F030 controller [42] that configures the transmission parameters. The LoRa node transmits signals at the 433 MHz frequency band, and the transmission power of the LoRa node is set to 20 dBm. Each LoRa gateway is built with two USRP B210 SDRs [49], which are synchronized by the SI5338 clock generator [40] to support four antennas. The distance between adjacent antennas is half the wavelength of the LoRa signal. The sampling rate of the gateway is set to 1 MHz.

We evaluate CLoc in both indoor and outdoor scenarios, as shown in Fig. 15. The outdoor environment is a 293,250 m^2 park with stands, fences, greenery, and occasional pedestrian activities (e.g., picnicking and running). The indoor environment is a 6,750 m^2 enclosed hall with chairs, tables, and walls. In each scenario, we deploy 30 LoRa nodes and 4 gateways. Each LoRa node's chirp is configured with specific Spreading Factors (SF) and Bandwidth (BW). SF varies from 6 to 12, and BW values are 125 kHz, 250 kHz, and 500 kHz. To generate concurrent collisions from multiple nodes, we use a beacon node to coordinate transmissions. Specifically, to generate a collision of N overlapping packets, the beacon node requests the transmission of N different LoRa nodes. By adjusting the number of transmitting nodes, we can conduct experiments at various levels of concurrency.

Metrics and groundtruth. The localization error is calculated as the absolute distance between the estimated location and the groundtruth. For the outdoor scenarios, we record the GPS locations of each node and gateway



(a) Effect of addressing inter-slope interference (b) Effect of addressing co-slope interference (c) Effect of addressing hybrid interference

Fig. 16. CSI measurements performance under different types of interference.

by the GPS Tracking APP on a smartphone. Additionally, we use the Xunta AK-800 laser rangefinder to refine these measurements and get the groundtruth. For the indoor scenarios, we employ the laser rangefinder to obtain the groundtruth of each node.

Comparison. Since CLoc focuses on extracting CSI in multi-target scenarios for localization, we compare CLoc with two baselines with different CSI extraction methods.

(1) *Seirios* [26]. Different from CLoc, Seirios estimates CSI without leveraging the LoRa chirp property and without resolving the interference among LoRa nodes. While Seirios adopts an advanced bandwidth aggregation technique to improve AoA estimation performance, it is designed for single-target scenarios and can not deal with the interference issue in multi-target scenarios. For a fair comparison, we use the same bandwidth for both Seirios and CLoc when estimating AoA.

(2) *CLoc-single.* This is a simplified version of CLoc. CLoc-single does not include CLoc's solutions to address the chirp interference and only works for a single node. CLoc-single adopts the CSI extraction method in Section 2.2 and CLoc 's joint localization. The purpose of comparing CLoc-single with CLoc is to demonstrate the importance of addressing chirp interference and the effectiveness of our interference removal method.

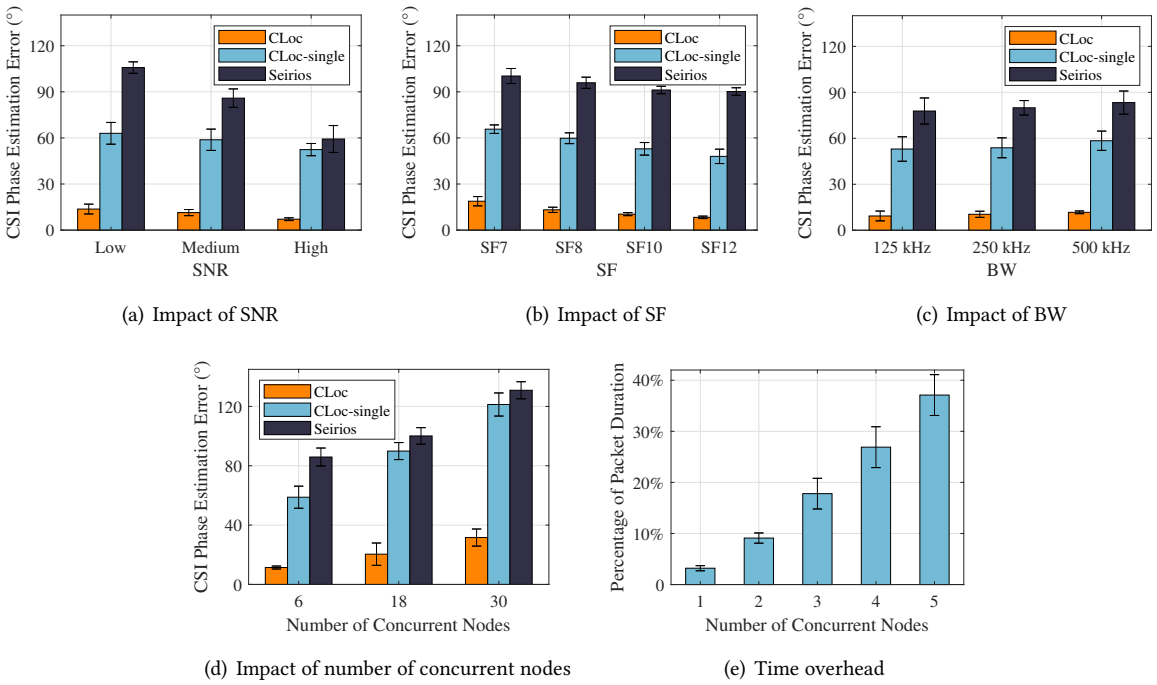
7.2 Performance of CSI Extraction in the Presence of Interference

In this section, we evaluate the performance of CLoc in multi-target CSI extraction. Given that AoA-based localization is primarily influenced by the accuracy of the CSI phase, we focus on the CSI phase estimation error as the key metric. We measure the CSI phase without interference as the groundtruth.

7.2.1 Performance under different types of interference. In this section, we first evaluate the effectiveness of CLoc on addressing different types of interference. For each experiment, we randomly select six nodes in the outdoor scenario to concurrently transmit chirp signals to the gateways. By controlling the chirp parameter (i.e., SF and BW) configuration of each node, we can generate two types of interference.

Effect of addressing inter-slope interference. In this experiment, we set the SF of six nodes to 7 to 12, respectively, and the BW is set to 500 kHz so that the transmitted chirps from different nodes have distinct chirp slopes. Fig. 16(a) illustrates the CSI phase estimation error in the presence of inter-slope interference. The median CSI phase estimation errors of CLoc-single and Seirios are 44.9° and 78.3°, respectively, indicating that the inter-slope interference can severely distort CSI. By leveraging the LoRa chirp property and resolving the interference among different LoRa nodes, CLoc decreases the median CSI phase estimation error to 11.8°.

Effect of addressing co-slope interference. In this experiment, we set the SF and BW of the six nodes to 8 and 500 kHz, respectively, so that the transmitted chirps from these nodes share the same slopes. Fig. 16(b) shows the CSI phase estimation error in the presence of co-slope interference. Seirios has a median CSI estimation error of around 79.1°, which is larger compared to CLoc-single and CLoc. Due to the CSI ambiguity and spectral



(a) Impact of SNR (b) Impact of SF (c) Impact of BW

(d) Impact of number of concurrent nodes (e) Time overhead

Fig. 17. CSI measurement performance under different settings.

leakage caused by the same slope interference, CLoc-single has a median CSI phase estimation error of 38.5°. CLoc further reduces the median error to just 8.3°, demonstrating its ability to address co-slope interference.

Effect of addressing hybrid interference. In practical scenarios, both inter-slope and co-slope interference can occur at the same time. Therefore, we also evaluate the performance of CLoc when the two types of interference occur simultaneously. We set the (SF, BW) configuration of three nodes to (7, 500 kHz), (9, 500 kHz), (10, 500 kHz) and the other three nodes to (8, 500 kHz) to generate hybrid interference. Fig. 16(c) shows the CSI phase estimation error in the presence of hybrid interference. The median CSI phase estimation error of Seirios is 78.6°. CLoc reduces this error to 9.7°, achieving an 87.7% improvement and highlighting its effectiveness in hybrid interference scenarios.

7.2.2 Performance under different settings. In this section, we evaluate the performance of CLoc under different signal-to-noise ratios (SNR), spreading factors (SF), bandwidths (BW), and number of concurrent nodes. First, we use six nodes to generate concurrent collisions and test the performance under different SNRs and SFs. Without loss of generality, the (SF, BW) configurations of the six nodes are set to (7, 500 kHz), (9, 500 kHz), (10, 500 kHz), (8, 500 kHz), (8, 500 kHz), (8, 500 kHz), respectively. Then, we increase the number of nodes to evaluate the scalability of CLoc.

Impact of SNR. SNR is one of the key factors affecting CSI extraction. In this experiment, we evaluate how SNR affects the performance of CSI estimation. By changing the transmission power of nodes, we evaluate the performance at different SNR levels, i.e., High (> 5 dB), Medium (5 dB ~ -5 dB), and Low (< -5 dB). Fig. 17(a) compares the performance of three methods under different SNR levels. Seirios, which does not utilize the chirp property, achieves similar performance to CLoc-single at the high SNR level, but its CSI phase estimation error increases to 105.8° in low-SNR scenarios. CLoc-single utilizes the LoRa chirp property which enables it to extract

CSI in low-SNR scenarios, and it achieves an error of 63.0° at low SNR level. However, this error is still large because CLoc-single is still severely affected by interference. CLoc further mitigates interference and performs well under different SNR levels. Even with low SNR, the CSI estimation error is just 13.7°.

Impact of SF. The spreading factor (SF) is a crucial LoRa chirp parameter that determines both the chirp duration and the ability of the chirp to combat noise. We evaluate the performance of CSI estimation under different SFs. In this experiment, the SNR is approximately between -5 dB~5 dB. As shown in Fig. 17(b), CLoc achieves much smaller CSI phase estimation errors, and the performance improves with the increase of SF. Specifically, when SF is set as 7, the CSI phase estimation error of CLoc is 18.8°. This error is significantly reduced to 8.3° when the SF is increased to 12. In contrast, Seirios has errors of 99.7° at SF = 7 and 90.2° at SF = 12.

Impact of BW. Bandwidth (BW) is another critical LoRa chirp parameter, determining the chirp duration and its capacity to combat noise. LoRa standard supports three typical bandwidth configurations, i.e., 125 kHz, 250 kHz, and 500 kHz. In this experiment, we evaluate the performance of CSI measurement under these different bandwidth settings. As shown in Fig. 17(c), the CSI phase estimation errors of CLoc at 125 kHz, 250 kHz, and 500 kHz are 9.3°, 10.4°, 11.7°, respectively. The narrower bandwidths generally lead to smaller CSI phase estimation errors because the narrower bandwidths usually result in longer chirp durations, which can provide better resistance to noise.

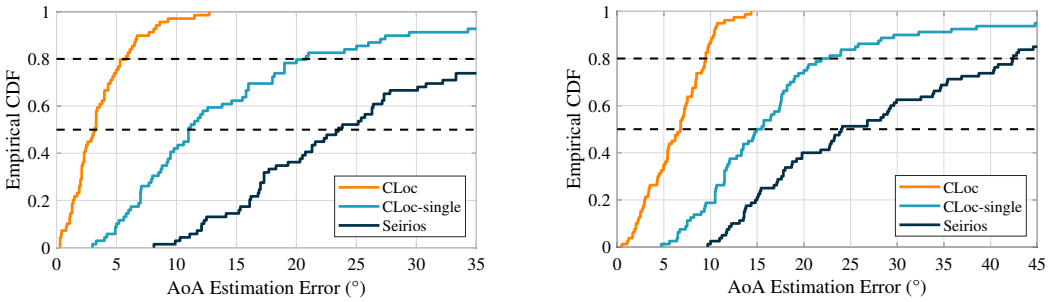
Impact of number of concurrent nodes. Now we evaluate the scalability of CLoc's CSI estimation accuracy by increasing the number of concurrent LoRa nodes. In this experiment, we increase the number of nodes from 6 to 30 to evaluate the impact on CSI estimation accuracy. As shown in Fig. 17(d), when the number of concurrent nodes increases, the performance of CLoc-single degrades significantly due to the lack of addressing the interference. In contrast, CLoc achieves much smaller CSI phase estimation errors. For example, when the number of nodes is 18, CLoc-single shows a CSI phase estimation error of 89.9°. The error of CLoc is just 20.1° even with 18 nodes transmitting signal concurrently. These results highlight the scalability of CLoc and its ability to maintain accurate CSI measurements even in high-density scenarios.

Computational overhead. In this experiment, we evaluate the computational overhead of CSI extraction in CLoc by measuring the time it takes to extract CSI from all nodes under varying numbers of concurrent transmissions. In the single-node scenario, the running time of CLoc is comparable to that required by the standard LoRa protocol to demodulate a single packet. This provides a baseline for evaluating the additional overhead introduced when handling multiple nodes. As the number of concurrent nodes increases, the overhead of CSI extraction in CLoc grows approximately linearly. As shown in Fig. 17(e), with two concurrent nodes, the overhead is about 2.9 times that of a single node. Even with five concurrent nodes, we find the overhead remains below 40% of one packet duration. This level of overhead is acceptable for the gateway, as it typically has higher computational capability and power supply than a normal LoRa node [54, 64].

7.3 Performance of AoA Estimation

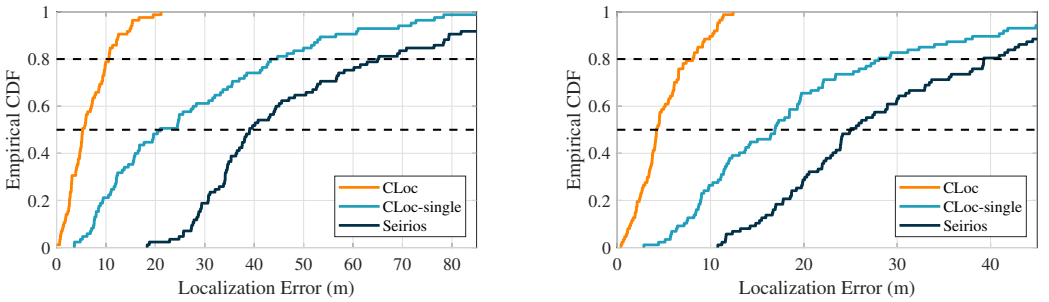
In this section, we evaluate the performance of CLoc in estimating the angle of arrival (AoA) in both indoor and outdoor environments, which is crucial for accurate localization. For each experiment, we randomly select four nodes from our deployment to transmit signals simultaneously. The parameters (SF, BW) of these four nodes are configured as (8, 500 kHz), (8, 500 kHz), (9, 500 kHz), and (10, 500 kHz). The gateway receives the signals of concurrent packets and estimates the AoA for each node.

Fig. 18 illustrates the AoA estimation errors in two different environments. In the outdoor scenario, as shown in Fig. 18(a), CLoc achieves a median AoA estimation error of 3.8°, significantly outperforming CLoc-single and Seirios, which have median AoA errors of 12.6° and 23.4°, respectively. This improvement highlights the effectiveness of CLoc to address the interference. In the indoor scenario, as shown in Fig. 18(b), larger AoA estimation errors occur due to the complex multipath propagation and channel fading. The median AoA estimation



(a) AoA estimation in outdoor scenario (b) AoA estimation in indoor scenario

Fig. 18. AoA estimation performance in outdoor and indoor scenarios.



(a) Localization in outdoor scenario (b) Localization in indoor scenario

Fig. 19. Localization performance in outdoor and indoor scenarios.

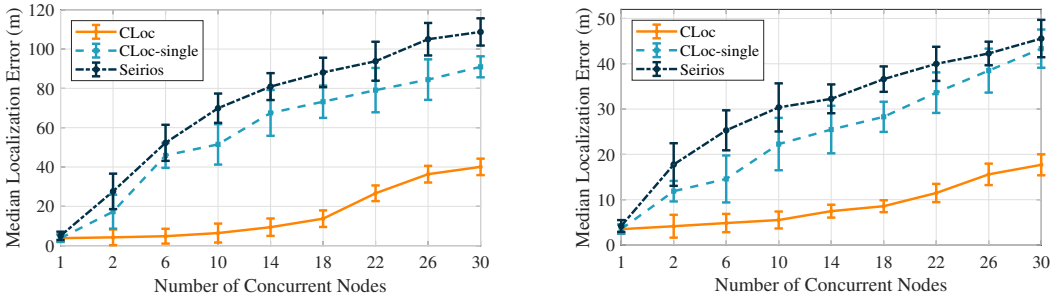
errors for CLoc, CLoc-single and Seirios are 6.1°, 15.4° and 24.8°, respectively. Despite in the complex indoor environment, CLoc still achieves good performance. Note that the performance in the outdoor environment is generally expected to be better than that of the indoor scenario because of less complex multipath conditions. However, data collection in the outdoor environment tends to be more heavily affected by other sources of interference, such as moving pedestrians and the random motion of trees and grass caused by wind.

7.4 Performance of Multi-target Localization

Now we evaluate the localization performance of CLoc under different settings. Since the AoA-based localization directly depends on the accuracy of CSI measurements and AoA estimations. The interference mitigation of CLoc plays a pivotal role in enhancing localization accuracy in both indoor and outdoor scenarios. Without loss of generality, we keep the same experimental setup as described in Section 7.3.

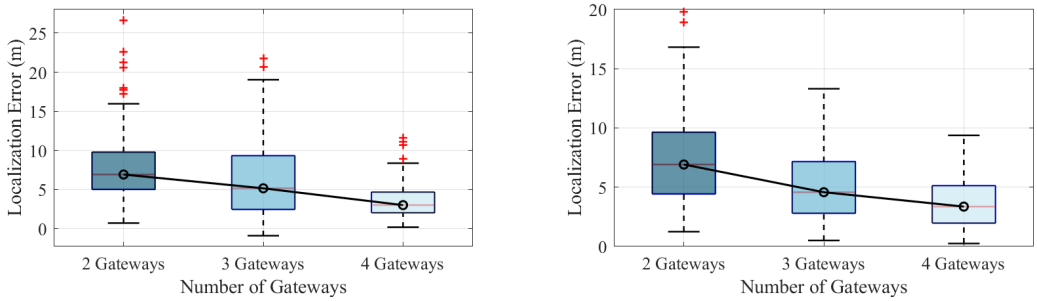
7.4.1 Localization accuracy. In this experiment, we select three gateways to localize the target for both indoor and outdoor environments. Note that the impacts of the number of LoRa nodes and the number of gateways are discussed in Section 7.4.2.

For the outdoor scenario, the overall performance of CLoc and the other two baselines is shown in Fig. 19(a). Seirios cannot effectively combat noise and interference for long-range localization, and the median localization



(a) Localization performance in outdoor scenario (b) Localization performance in indoor scenario

Fig. 20. Impact of the number of LoRa nodes.



(a) Localization performance in outdoor scenario (b) Localization performance in indoor scenario

Fig. 21. Impact of the number of LoRa gateways.

error is 39.8 m. The median localization error of CLoc-single is 25.6 m, which assumes the signal is interference-free. The concurrent transmissions introduce interference that degrades CSI estimation accuracy, thus degrading localization performance. CLoc achieves a median localization error of 4.9 m, outperforming CLoc-single and Seirios. Overall, CLoc achieves a median localization error that is 80.9% and 87.7% smaller than that of CLoc-single and Seirios in multi-target scenarios.

For the indoor scenario, the corresponding localization performance is presented in Fig. 19(b). The median localization error of CLoc is 4.3 m, while the errors of CLoc-single and Seirios are 17.5 m and 25.4 m, respectively. Similar to the outdoor scenario, CLoc-single and Seirios are severely affected by concurrent interference, resulting in significant performance degradation. The localization performance of CLoc is 83.1% better than Seirios.

7.4.2 Performance under different network scales. In this section, we evaluate the localization performance of CLoc under different network scales, i.e., varying the number of LoRa nodes and the number of gateways.

Impact of the number of LoRa nodes. We increase the number of concurrent nodes from 1 to 30 to evaluate the localization performance of CLoc. Fig. 20(a) and Fig. 20(b) show the experiment results for outdoor and indoor scenarios, respectively. When there is only one LoRa node, the performance of CLoc-single is the same as that of CLoc. Seirios estimates CSI without using the LoRa chirp property and does not address interference among LoRa nodes. In the absence of interference, the localization error of Seirios is larger than that of CLoc-single, due to its poor CSI estimation performance under low-SNR conditions. In the outdoor scenario, Seirios and CLoc achieve median localization errors of 5.1 m and 4.1 m, respectively. In the indoor scenario, the median localization errors

of Seirios and CLoc are 4.4 m and 3.9 m, respectively. As the number of nodes increases, the median localization errors for Seirios and CLoc-single increase significantly, whereas the errors of CLoc increase more steadily. With 10 concurrent nodes, CLoc achieves a median localization error of 6.5 m in the outdoor environment, which has a 90.6% reduction compared to that of Seirios. In the indoor scenario, CLoc achieves a median localization error of 5.8 m, reducing the localization error by 80.0% compared to Seirios.

Impact of the number of LoRa gateways. Multiple gateways are employed to estimate target locations. We evaluate the impact of the number of gateways in both outdoor and indoor scenarios. In this experiment, we randomly choose four nodes to transmit signals simultaneously. Fig. 21(a) and Fig. 21(b) demonstrate the localization errors in two different scenarios. With just two gateways, CLoc achieves a median localization error of 7.3 m in the outdoor scenario and 7.1 m in the indoor scenario, respectively. Increasing the number of gateways to three greatly reduces the localization errors to 4.9 m in outdoor environments and 4.3 m in indoor environments. Further increasing to four gateways results in greater improvements, with errors decreasing to 3.3 m and 3.5 m, respectively. More gateways provide more information for triangulation-based location estimation, thus improving the overall localization accuracy. However, there is a trade-off between localization performance and deployment cost.

8 DISCUSSION

In this work, we focus on addressing different types of interference and achieving multi-target LoRa localization. We briefly discuss the limitations and future direction of our work below.

Mitigating cross-technology interference. Besides the interference caused by concurrent transmissions within the LoRa network, LoRa localization is also affected by Cross-Technology Interference (CTI) from other wireless technologies operating in the same frequency band. In LoRa communication, the impact of these heterogeneous interference can be effectively alleviated by detecting and eliminating them in the time-frequency domain [44, 58]. We believe combining these methods and CLoc can mitigate the impact of heterogeneous interference on LoRa localization.

Localizing more LoRa nodes. Our design addresses interference in multi-target scenarios, enabling accurate localization of multiple LoRa nodes. However, when the number of concurrent nodes is very large (e.g., 40), our system experiences significant localization errors. In extreme collision scenarios, the whole data packets could be overwhelmed by interference, making it more challenging to extract CSI. We plan to apply deep learning to deal with this challenging problem.

Comparing with MAC-layer scheduling. MAC-layer scheduling methods, especially TDMA (Time Division Multiple Access), can avoid conflicts and thus address the interference. However, TDMA induces large latency and computational overhead due to time allocation, time slot management, and the need for strict synchronization. We believe that combining our method with MAC-layer scheduling techniques can more effectively reduce interference in multi-target LoRa localization.

Hardware cost. In our experiments, each LoRa gateway consists of two USRP B210 boards [49] and one SI5338 clock generator [40]. Each USRP B210 costs $2165, while the SI5338 clock generator is priced at $59. To reduce hardware cost, we plan to employ cheaper SDRs, such as BladeRF [7], LimeSDR [24], and KrakenSDR [18], or the commercial LoRa gateway equipped with the SX1303 chip [52] in the future. While these budget-friendly alternatives can help reduce costs, they often come with trade-offs in certain performance metrics, such as lower analog-to-digital converter (ADC) resolution. For example, the KrakenSDR has an ADC resolution that is 4 bits lower than that of the USRP, indicating that it provides less precision in capturing signal amplitudes. This reduction in resolution can lead to decreased receiver sensitivity and a shorter sensing range. Therefore, it is crucial to select cost-effective hardware that still meets the necessary performance requirements.

9 RELATED WORK

9.1 Wireless Localization

Localization has been intensively studied in the past few decades. Global Positioning System (GPS) [43] offers pervasive localization for outdoor scenarios but fails in indoor settings. In addition, it requires extensive energy consumption and is not friendly to low-power devices with limited batteries. To mitigate these issues, researchers have studied to employ wireless technologies for localization, such as WiFi [17, 56, 65], ZigBee [1, 21], Bluetooth [4, 11, 66], and UWB (Ultra Wide Bandwidth) [50, 60]. However, these techniques commonly focus on short-range localization (e.g., 30 m) that inevitably leads to heavy infrastructure deployment for a large-scale scenario.

9.2 LoRa Localization

In the area of LoRa localization, the early studies rely on the Received Signal Strength Indicator (RSSI) [2, 19, 20, 25], Time of Flight (ToF) [31], or Time Difference of Arrival (TDoA) [5, 15, 16]. Although these methods demonstrate the feasibility of LoRa-based localization, they commonly suffer from significant localization errors.

Traditional RSSI-based methods, due to the inherently coarse nature of RSSI and LoRa's narrow bandwidth (e.g., 125 kHz), typically achieve an accuracy ranging from 500 m to 2,000 m [27]. The limited bandwidth of LoRa also constrains ToF-based solutions. Although the latest 2.4 GHz SX1280 LoRa chip supports a maximum bandwidth of 1625 kHz [37], this bandwidth results in a range resolution of 92 m, which is still insufficient for precise localization [13]. The commercial 2.4 GHz LoRa positioning system [31] uses the ToF-based ranging engine provided by the SX1280 chip [37] to calculate distances from the target node to multiple anchor nodes, and then calculates the location of the target node, achieving a median localization error of 30 m.

For the TDoA-based methods, OwLL [6] and ILLOC [13] are two recent representative methods. OwLL [6] uses TV white spaces for frequency hopping to overcome the narrowband limitation of LoRa, achieving a median localization error of 9 m. However, it operates across a broad 400 MHz frequency range, exceeding the LoRa standard frequency band (902-928 MHz) [41], and requires compliance with regional TV white space policies. Moreover, OwLL needs to send 80 to 120 packets per localization query, leading to huge time and energy consumption. ILLOC [13] improves time resolution through a high sampling rate (i.e., 25 Msps), but the range resolution it provides is still limited to 12 m, and further increasing the sampling rate increases the performance and cost requirements of the gateway. In contrast, CLoc relies on AoA estimation for localization, which is not limited by the bandwidth of LoRa and does not require a high sampling rate.

To further enable accurate localization, recent advances turn to calculating CSI since CSI offers a comprehensive representation of the channel properties between the LoRa node and gateway. Based on CSI, recent studies propose angle-of-arrival (AoA)-based localization methods and fingerprinting-based localization. For AoA-based methods [8, 9, 26], it calculates the CSI of received signals and estimates AoA by using a super-resolution algorithm [33, 34] via multi-antenna gateways. For CSI fingerprinting-based localization, OrchLoc [61] explores the CSI-based fingerprint localization method, combined with deep learning, to locate trees in orchards. However, it relies on a well-collected fingerprint database, which can be time-consuming to create and can not generalize well to different environments. Despite these efforts, these CSI-based localization methods assume that received signals are free of interference and there is only one LoRa node transmitting signals. This is unlikely in practice, given the long communication range and wide deployment of LoRa. To address this issue, CLoc offers an AoA-based localization technique that addresses the interference caused by concurrent LoRa transmissions to support multi-target localization.

9.3 Concurrent LoRa Communication

Concurrent LoRa communication has been extensively studied by existing literature [10, 12, 14, 22, 23, 39, 46, 48, 51, 54, 59, 62, 63]. However, concurrent LoRa communication and concurrent LoRa localization are

fundamentally different research topics with their own unique challenges. The concurrent LoRa communication typically addresses the frequency ambiguity issue caused by co-slope interference, while the impact of inter-slope interference is negligible in the field of communication. In contrast, the inter-slope interference can cause severe CSI distortion in LoRa localization. CLoc designs novel methods to address these types of interference.

10 CONCLUSION

In this paper, we investigate the chirp interference in LoRa localization and present CLoc, the first LoRa-based multi-target localization method to address different types of chirp interference. Specifically, by carefully addressing the inter-slope interference and co-slope interference caused by concurrent LoRa transmissions, CLoc extracts precise CSI and offers accurate localization of multiple LoRa nodes simultaneously. Comprehensive experiments in different scenarios validate the effectiveness of CLoc in multi-target LoRa localization, demonstrating its potential in large-scale IoT applications.

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